Koalified build order

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Build version 1 in six phases, one screen group at a time, starting with fake data and adding storage once the screens work.

Data models

Six SwiftData types cover every screen in the Figma file. Build them in this order, because each one depends on the one above it.

Type	Holds	Used by
Semester	Name, start and end dates, break weeks, last day to drop	Calendar, Today
Course	Code, name, semester, hurdle on or off, hurdle cutoff (default 40%)	Courses, Calculator, Degree plan
Assessment	Name, weight, due date, mark (empty until entered), course	Course detail, Calculator, Today
ClassEvent	Course, weekday, start and end time, room, repeat weeks	Today agenda, Class details
Goal	Title, target, progress	Goals, Today
ImportRecord	Source (.ics, Excel, pasted text, manual), date, what was wrong	Import check

The grade logic is plain Swift functions, not stored data. Write these first and test them with a few real courses:

- running mark = marks earned so far divided by weight marked so far
- needed mark = (target total minus points banked minus assumed points) divided by the weight left
- grade bands: 4 from 50, 5 from 65, 6 from 75, 7 from 85
- hurdle check: a failed hurdle overrides the grade

Build order

Each phase ends with something you can run on your own phone. Learn only the topic in the last column before starting the phase.

Phase	Figma screens	SwiftUI pieces	Learn first
1. Shell and fake data	Tab bar, Today, Courses	TabView, NavigationStack, List, hard-coded sample data	Apple SwiftUI Essentials tutorial
2. Course and marks	Course detail, enter mark sheet, Set up course	Sheets, TextField, Toggle, forms, state and bindings	Hacking with Swift: state and forms
3. Calculator	Calculator, try different marks sheet, hurdle states	Picker or buttons for grade, Slider with a number field, computed values	Plain Swift functions and unit tests
4. Storage	All of the above	SwiftData models, @Query, saving and editing	WWDC session on SwiftData
5. Import	Import timetable, import check, Excel import	fileImporter, parsing .ics text, review list	Swift file reading and string parsing
6. Calendar, help, widgets	Semester calendar, Help and FAQ, widgets, reminders	LazyVGrid for the month, WidgetKit with an App Group, local notifications	WWDC sessions on WidgetKit and notifications

Do onboarding, the tips, profile and Degree plan last, because they only wrap the screens above. Phase 3 can start before phase 4, because the calculator only needs numbers.

Shared components

Build these once in phase 1 and reuse them. They match the design system in Figma.

Component	Where it appears	Notes
Card	Today, Course detail, calculator	White, 14 pt radius, light border and shadow
Agenda row	Today	Time, title, detail, optional Urgent badge with a clock icon
Assessment row	Course detail	Whole row is the tap target, minimum 44 pt tall

Component	Where it appears	Notes
Status badge	Urgent, Verified, Unverified, Hurdle	Always an icon or word as well as colour
Segmented switch	Agenda and Calendar, My courses and Degree plan	Native Picker with the segmented style
Bottom sheet	Enter mark, try different marks, add a deadline	.sheet with presentation detents
Grade pills	Calculator	Four buttons, one selected
Number and slider row	Try different marks	Slider and TextField bound to one value

Use the Inter font only if you add the font files to the project. Otherwise the system font (SF Pro) is the safe default and supports Dynamic Type automatically.

Version roadmap

Each version has one goal and a clear finish line. Do not start the next version until the current one is done. The dates are targets, not promises, and they leave room for your exams (COMP3506 on 11 Nov, CSSE3200 on 17 Nov).

Version	Goal	What ships	Done when
v0.1 Prototype (Nov to Dec 2026)	Prove the core works on your own phone	Build phases 1 to 4: tab shell, Today, Courses, course detail, enter marks, calculator, SwiftData storage	You use it for your own real courses for one week without a crash, and the grade maths passes unit tests on 3 real courses
v0.5 Beta (Jan 2027)	Get it ready for other people	Build phases 5 and 6: timetable and Excel import with the import check, semester calendar, widgets, reminders, onboarding with sample data, tips, Help and FAQ	10 friends install it from TestFlight and 8 of them finish setup without your help

Version	Goal	What ships	Done when
v1.0 App Store launch (target: before Semester 1 starts, late Feb 2027)	Launch to UQ students	Everything in v0.5, polished: accessibility pass, animations and haptics, privacy policy, App Store listing and screenshots	App is approved and live, with no sign-up, no backend and no UQ login
v1.1 Fixes and feedback (Mar 2027)	Fix what real students hit	Bug fixes, import fixes for timetable formats you missed, small wording and layout changes	100 installs, crash-free rate above 99%, and the top 5 complaints are fixed
v1.2 Growth (Apr to May 2027)	Make students tell their friends	Share your grade card as an image, Degree plan, dark mode, optional course-profile search if UQ has given written permission	100 weekly active users and students still opening it in week 6
v2.0 Accounts and sync (Semester 2 2027)	Let students move between devices	Sign-up and login, sync across devices, student-verified deadlines, profile sharing	Students can sign in on a second phone and see the same courses, and the privacy and security review is done
v3.0 Plus and social (late 2027 onward)	Earn money and keep students coming back	Plus subscription, study coach and planner, GPA forecast, Momentum streaks, study buddies and Shared Fires	Paying users, and you have pitched to UQ Ventures

Rules for the roadmap

- Anything not listed in v0.1 to v1.0 waits. If you are tempted to add it, write it in the next version instead.
- Before v1.0, check the policy items in the architecture doc: no UQ passwords, no scraping behind login, and written permission before reading any UQ page.
- Keep the Apple Developer account, app name and koala artwork rights sorted before v0.5, because TestFlight needs them.

- Take the pitch to UQ Ventures after v1.1 or v1.2, once you can show real user numbers.
- If a version slips, move the dates and keep the order.

Left for later

These are in Figma under Upcoming. Do not build them for version 1.

- Sign-up, login and sync across devices
- Study coach, smart study planner and the GPA forecast (Plus)
- Momentum streaks, study buddies and Shared Fires
- Profile sharing and student-verified deadlines
- Reading course profiles from a link or course code (needs UQ's written permission first)
- Dark mode (the contrast table is ready when you want it)

Open question: whether to build in SwiftUI only for iPhone. The Figma screens are 393 pt wide, so iPhone-only is the simplest start.